

Bounds for Integrals. ML-Inequality

There will be a frequent need for estimating the absolute value of complex line integrals

The basic formula is

$$\left| \int_C f(z) dz \right| \leq ML \quad (\text{ML-Inequality})$$

where L is the length of C and M a constant such that $|f(z)| \leq M$ everywhere on C .

Example Estimation of an Integral.

Find an upper bound for the absolute value of the integral

$$\int_C z^2 dz$$

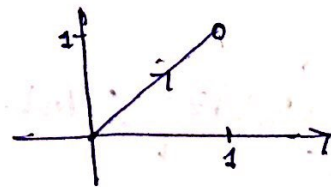
$\because C$ is the straight-line segment from 0 to $1+i$

Solution:-

As we know that

$$\left| \int_C f(z) dz \right| \leq ML \rightarrow (1)$$

where $L = \int_a^b |z'(t)| dt$, $M = |f(z)|$



Path in Example.

$$z(t) = x(t) + iy(t)$$

$$z(t) = t + it$$

As

$$0 \leq x \leq 1$$

$$0 \leq t \leq 1$$

$$\begin{matrix} x_1 & y_1 & x_2 & y_2 \\ (0, 0) & \rightarrow & (1, 1) \end{matrix}$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$y = \frac{1-0}{1-0} (x-0)$$

$$y = x$$

let
 $x = t$

$$y = t$$

$$z'(t) = 1 + i$$

$$|z'(t)| = |1+i| = \sqrt{1+1} = \sqrt{2}$$

$$L = \int_a^b |z'(t)| dt = \sqrt{2} \int_0^1 (1) dt = \sqrt{2}$$

$$L = \sqrt{2}$$

$$f(z) = z^2$$

where

$$z(t) = t + it$$

$$M = |f(z)| = |t + it|^2$$

$$\Rightarrow |(1+i)|^2 = |(1+i^2 + 2i)|$$

$$= |1 - 1 + 2i| = |2i|$$

$$\Rightarrow \sqrt{0^2 + 2^2} = 2$$

max value of
 $0 \leq t \leq 1$

from eq ① we have.

$$\left| \int_C z^2 dz \right| \leq ML$$

$$\Rightarrow \left| \int_C z^2 dz \right| \leq 2\sqrt{2} = 2.8284 \rightarrow \text{Answer}$$

L.H.S

The absolute value of The Integral

$$\left| \int_0^{1+i} z^2 dz \right| = ??$$

$$\left| \int_0^{1+i} z^2 dz \right| \rightarrow \textcircled{A}$$

Consider $\int_0^{1+i} z^2 dz$.

As $f(z) = z^2$ which is analytic so By the 1st evaluation method.

$$\int_0^{1+i} z^2 dz = \left. \frac{z^3}{3} \right|_0^{1+i} = \frac{(1+i)^3}{3} = -\frac{2}{3} + \frac{2}{3}i$$

So from $\textcircled{A}$ we have.

$$\left| -\frac{2}{3} + \frac{2}{3}i \right| = \frac{2}{3}\sqrt{2} = 0.9428$$

From eq $\textcircled{*}$ we have.

$$\boxed{0.9428 \leq 2.8284}$$

Summary on Integration:-

Line integrals of $f(z)$ can always be evaluated by Theorem (2) using a representation $z(t) = x(t) + iy(t)$ of the path of integration.

$\textcircled{*}$ If $f(z)$ is analytic, Indefinite Integration by (Theorem 4) as in calculus will be simpler.

Recommended Ex# 14.1

14.2 Cauchy's Integral Theorem

This section is the focal point of the chapter. We have just seen in Sec. 14.1 that a line integral of a function $f(z)$ generally depends not merely on the endpoints of the path, but also on the choice of the path itself. This dependence often complicates situations. Hence conditions under which this does *not* occur are of considerable importance. Namely, if $f(z)$ is analytic in a domain D and D is simply connected (see Sec. 14.1 and also below), then the integral will not depend on the choice of a path between given points. This result (Theorem 2) follows from Cauchy's integral theorem, along with other basic consequences that make *Cauchy's integral theorem the most important theorem in this chapter* and fundamental throughout complex analysis.

Let us continue our discussion of simple connectedness which we started in Sec. 14.1.

1. A **simple closed path** is a closed path (defined in Sec. 14.1) that does not intersect or touch itself as shown in Fig. 345. For example, a circle is simple, but a curve shaped like an 8 is not simple.

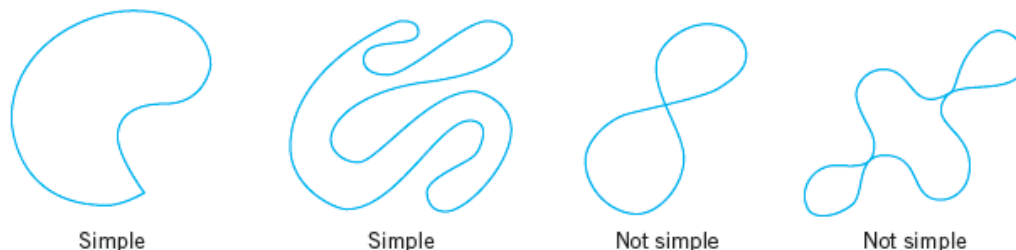


Fig. 345. Closed paths

2. A **simply connected domain** D in the complex plane is a domain (Sec. 13.3) such that every simple closed path in D encloses only points of D . *Examples:* The interior of a circle ("open disk"), ellipse, or any simple closed curve. A domain that is not simply connected is called **multiply connected**. *Examples:* An annulus (Sec. 13.3), a disk without the center, for example, $0 < |z| < 1$. See also Fig. 346.

More precisely, a **bounded domain** D (that is, a domain that lies entirely in some circle about the origin) is called **p -fold connected** if its boundary consists of p closed

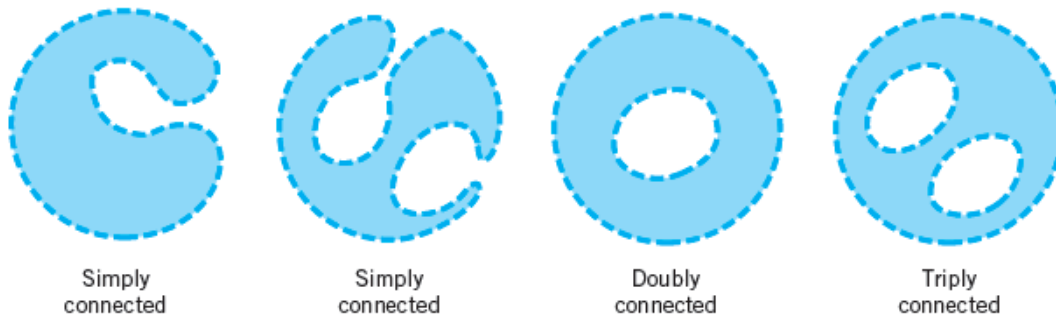


Fig. 346. Simply and multiply connected domains

connected sets without common points. These sets can be curves, segments, or single points (such as $z = 0$ for $0 < |z| < 1$, for which $p = 2$). Thus, D has $p - 1$ “holes,” where “hole” may also mean a segment or even a single point. Hence an annulus is doubly connected ($p = 2$).

THEOREM 1

Cauchy's Integral Theorem

If $f(z)$ is analytic in a simply connected domain D , then for every simple closed path C in D ,

(1)

$$\oint_C f(z) dz = 0.$$

See Fig. 347.

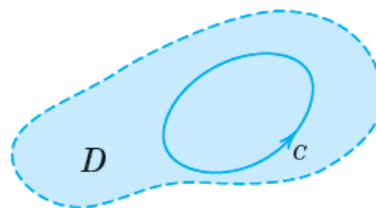


Fig. 347. Cauchy's integral theorem

Before we prove the theorem, let us consider some examples in order to really understand what is going on. A simple closed path is sometimes called a *contour* and an integral over such a path a **contour integral**. Thus, (1) and our examples involve contour integrals.

EXAMPLE 1 Entire Functions

$$\oint_C e^z dz = 0, \quad \oint_C \cos z dz = 0, \quad \oint_C z^n dz = 0 \quad (n = 0, 1, \dots)$$

for any closed path, since these functions are entire (analytic for all z).

EXAMPLE 2 Points Outside the Contour Where $f(z)$ is Not Analytic

$$\oint_C \sec z \, dz = 0, \quad \oint_C \frac{dz}{z^2 + 4} = 0$$

where C is the unit circle, $\sec z = 1/\cos z$ is not analytic at $z = \pm\pi/2, \pm3\pi/2, \dots$, but all these points lie outside C ; none lies on C or inside C . Similarly for the second integral, whose integrand is not analytic at $z = \pm 2i$ outside C . ■

EXAMPLE 3 Nonanalytic Function

$$\oint_C \bar{z} \, dz = \int_0^{2\pi} e^{-it} i e^{it} dt = 2\pi i$$

where $C: z(t) = e^{it}$ is the unit circle. This does not contradict Cauchy's theorem because $f(z) = \bar{z}$ is not analytic. ■

EXAMPLE 4 Analyticity Sufficient, Not Necessary

$$\oint_C \frac{dz}{z^2} = 0$$

where C is the unit circle. This result does *not* follow from Cauchy's theorem, because $f(z) = 1/z^2$ is not analytic at $z = 0$. Hence *the condition that f be analytic in D is sufficient rather than necessary for (1) to be true.* ■

EXAMPLE 5 Simple Connectedness Essential

$$\oint_C \frac{dz}{z} = 2\pi i$$

for counterclockwise integration around the unit circle (see Sec. 14.1). C lies in the annulus $\frac{1}{2} < |z| < \frac{3}{2}$ where $1/z$ is analytic, but this domain is not simply connected, so that Cauchy's theorem cannot be applied. Hence *the condition that the domain D be simply connected is essential.*

In other words, by Cauchy's theorem, if $f(z)$ is analytic on a simple closed path C and everywhere inside C , with no exception, not even a single point, then (1) holds. The point that causes trouble here is $z = 0$ where $1/z$ is not analytic. ■

PROOF Cauchy proved his integral theorem under the additional assumption that the derivative $f'(z)$ is continuous (which is true, but would need an extra proof). His proof proceeds as follows. From (8) in Sec. 14.1 we have

$$\oint_C f(z) \, dz = \oint_C (u \, dx - v \, dy) + i \oint_C (u \, dy + v \, dx).$$

Since $f(z)$ is analytic in D , its derivative $f'(z)$ exists in D . Since $f'(z)$ is assumed to be continuous, (4) and (5) in Sec. 13.4 imply that u and v have *continuous* partial derivatives in D . Hence Green's theorem (Sec. 10.4) (with u and $-v$ instead of F_1 and F_2) is applicable and gives

$$\oint_C (u \, dx - v \, dy) = \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx \, dy$$

where R is the region bounded by C . The second Cauchy–Riemann equation (Sec. 13.4) shows that the integrand on the right is identically zero. Hence the integral on the left is zero. In the same fashion it follows by the use of the first Cauchy–Riemann equation that the last integral in the above formula is zero. This completes Cauchy's proof. ■

Goursat's proof *without the condition that $f'(z)$ is continuous*¹ is much more complicated. We leave it optional and include it in App. 4.

Independence of Path

We know from the preceding section that the value of a line integral of a given function $f(z)$ from a point z_1 to a point z_2 will in general depend on the path C over which we integrate, not merely on z_1 and z_2 . It is important to characterize situations in which this difficulty of path dependence does not occur. This task suggests the following concept. We call an integral of $f(z)$ **independent of path in a domain D** if for every z_1, z_2 in D its value depends (besides on $f(z)$, of course) only on the initial point z_1 and the terminal point z_2 , but not on the choice of the path C in D [so that every path in D from z_1 to z_2 gives the same value of the integral of $f(z)$].

THEOREM 2

Independence of Path

If $f(z)$ is analytic in a simply connected domain D , then the integral of $f(z)$ is independent of path in D .

PROOF Let z_1 and z_2 be any points in D . Consider two paths C_1 and C_2 in D from z_1 to z_2 without further common points, as in Fig. 348. Denote by C_2^* the path C_2 with the orientation reversed (Fig. 349). Integrate from z_1 over C_1 to z_2 and over C_2^* back to z_1 . This is a simple closed path, and Cauchy's theorem applies under our assumptions of the present theorem and gives zero:

$$(2') \quad \int_{C_1} f \, dz + \int_{C_2^*} f \, dz = 0, \quad \text{thus} \quad \int_{C_1} f \, dz = - \int_{C_2^*} f \, dz.$$

But the minus sign on the right disappears if we integrate in the reverse direction, from z_1 to z_2 , which shows that the integrals of $f(z)$ over C_1 and C_2 are equal,

$$(2) \quad \int_{C_1} f(z) dz = \int_{C_2} f(z) dz \quad (\text{Fig. 348}).$$

This proves the theorem for paths that have only the endpoints in common. For paths that have finitely many further common points, apply the present argument to each “loop” (portions of C_1 and C_2 between consecutive common points; four loops in Fig. 350). For paths with infinitely many common points we would need additional argumentation not to be presented here.

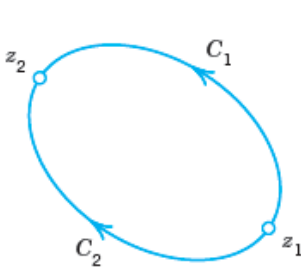


Fig. 348. Formula (2)

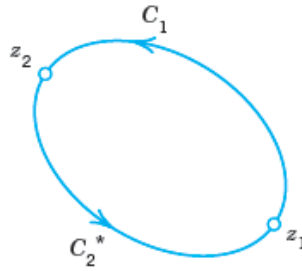


Fig. 349. Formula (2')

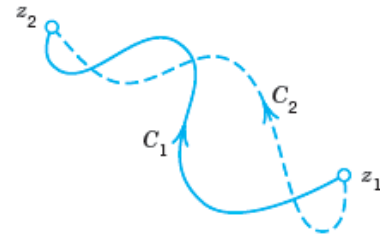


Fig. 350. Paths with more common points

Principle of Deformation of Path

This idea is related to path independence. We may imagine that the path C_2 in (2) was obtained from C_1 by continuously moving C_1 (with ends fixed!) until it coincides with C_2 . Figure 351 shows two of the infinitely many intermediate paths for which the integral always retains its value (because of Theorem 2). Hence we may impose a continuous deformation of the path of an integral, keeping the ends fixed. As long as our deforming path always contains only points at which $f(z)$ is analytic, the integral retains the same value. This is called the **principle of deformation of path**.

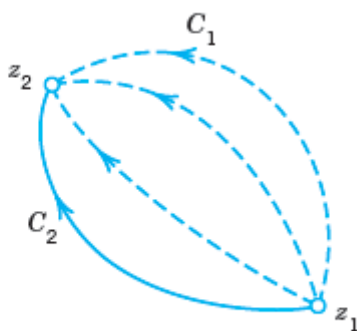


Fig. 351. Continuous deformation of path

EXAMPLE 6 A Basic Result: Integral of Integer Powers

From Example 6 in Sec. 14.1 and the principle of deformation of path it follows that

$$(3) \quad \oint (z - z_0)^m dz = \begin{cases} 2\pi i & (m = -1) \\ 0 & (m \neq -1 \text{ and integer}) \end{cases}$$

for counterclockwise integration around *any simple closed path containing z_0 in its interior*.

Indeed, the circle $|z - z_0| = \rho$ in Example 6 of Sec. 14.1 can be continuously deformed in two steps into a path as just indicated, namely, by first deforming, say, one semicircle and then the other one. (Make a sketch). ■

Existence of Indefinite Integral

We shall now justify our indefinite integration method in the preceding section [formula (9) in Sec. 14.1]. The proof will need Cauchy's integral theorem.

THEOREM 3

Existence of Indefinite Integral

If $f(z)$ is analytic in a simply connected domain D , then there exists an indefinite integral $F(z)$ of $f(z)$ in D —thus, $F'(z) = f(z)$ —which is analytic in D , and for all paths in D joining any two points z_0 and z_1 in D , the integral of $f(z)$ from z_0 to z_1 can be evaluated by formula (9) in Sec. 14.1.

PROOF The conditions of Cauchy's integral theorem are satisfied. Hence the line integral of $f(z)$ from any z_0 in D to any z in D is independent of path in D . We keep z_0 fixed. Then this integral becomes a function of z , call it $F(z)$,

$$(4) \quad F(z) = \int_{z_0}^z f(z^*) dz^*$$

which is uniquely determined. We show that this $F(z)$ is analytic in D and $F'(z) = f(z)$. The idea of doing this is as follows. Using (4) we form the difference quotient

$$(5) \quad \frac{F(z + \Delta z) - F(z)}{\Delta z} = \frac{1}{\Delta z} \left[\int_{z_0}^{z+\Delta z} f(z^*) dz^* - \int_{z_0}^z f(z^*) dz^* \right] = \frac{1}{\Delta z} \int_z^{z+\Delta z} f(z^*) dz^*.$$

We now subtract $f(z)$ from (5) and show that the resulting expression approaches zero as $\Delta z \rightarrow 0$. The details are as follows.

We keep z fixed. Then we choose $z + \Delta z$ in D so that the whole segment with endpoints z and $z + \Delta z$ is in D (Fig. 352). This can be done because D is a domain, hence it contains a neighborhood of z . We use this segment as the path of integration in (5). Now we subtract $f(z)$. This is a constant because z is kept fixed. Hence we can write

$$\int_z^{z+\Delta z} f(z) dz^* = f(z) \int_z^{z+\Delta z} dz^* = f(z) \Delta z. \quad \text{Thus} \quad f(z) = \frac{1}{\Delta z} \int_z^{z+\Delta z} f(z) dz^*.$$

By this trick and from (5) we get a single integral:

$$\frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) = \frac{1}{\Delta z} \int_z^{z+\Delta z} [f(z^*) - f(z)] dz^*.$$

Since $f(z)$ is analytic, it is continuous (see Team Project (24d) in Sec. 13.3). An $\epsilon > 0$ being given, we can thus find a $\delta > 0$ such that $|f(z^*) - f(z)| < \epsilon$ when $|z^* - z| < \delta$. Hence, letting $|\Delta z| < \delta$, we see that the *ML*-inequality (Sec. 14.1) yields

$$\left| \frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) \right| = \frac{1}{|\Delta z|} \left| \int_z^{z+\Delta z} [f(z^*) - f(z)] dz^* \right| \leq \frac{1}{|\Delta z|} \epsilon |\Delta z| = \epsilon.$$

By the definition of limit and derivative, this proves that

$$F'(z) = \lim_{\Delta z \rightarrow 0} \frac{F(z + \Delta z) - F(z)}{\Delta z} = f(z).$$

Since z is any point in D , this implies that $F(z)$ is analytic in D and is an indefinite integral or antiderivative of $f(z)$ in D , written

$$F(z) = \int f(z) dz.$$

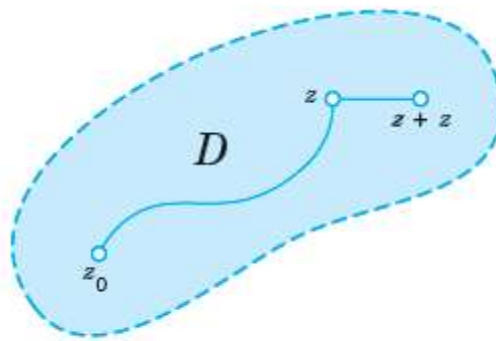


Fig. 352. Path of integration

Also, if $G'(z) = f(z)$, then $F'(z) - G'(z) \equiv 0$ in D ; hence $F(z) - G(z)$ is constant in D (see Team Project 30 in Problem Set 13.4). That is, two indefinite integrals of $f(z)$ can differ only by a constant. The latter drops out in (9) of Sec. 14.1, so that we can use any indefinite integral of $f(z)$. This proves Theorem 3. ■

Cauchy's Integral Theorem for Multiply Connected Domains

Cauchy's theorem applies to multiply connected domains. We first explain this for a **doubly connected domain** D with outer boundary curve C_1 and inner C_2 (Fig. 353). If a function $f(z)$ is analytic in any domain D^* that contains D and its boundary curves, we claim that

$$(6) \quad \oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz \quad (\text{Fig. 353})$$

both integrals being taken counterclockwise (or both clockwise, and regardless of whether or not the full interior of C_2 belongs to D^*).

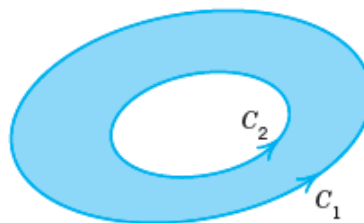


Fig. 353. Paths in (5)

PROOF By two cuts $\tilde{C}_1$ and $\tilde{C}_2$ (Fig. 354) we cut D into two simply connected domains D_1 and D_2 in which and on whose boundaries $f(z)$ is analytic. By Cauchy's integral theorem the integral over the entire boundary of D_1 (taken in the sense of the arrows in Fig. 354) is zero, and so is the integral over the boundary of D_2 , and thus their sum. In this sum the integrals over the cuts $\tilde{C}_1$ and $\tilde{C}_2$ cancel because we integrate over them in both directions—this is the key—and we are left with the integrals over C_1 (counterclockwise) and C_2 (clockwise; see Fig. 354); hence by reversing the integration over C_2 (to counterclockwise) we have

$$\oint_{C_1} f dz - \oint_{C_2} f dz = 0$$

and (6) follows. ■

For domains of higher connectivity the idea remains the same. Thus, for a **triply connected domain** we use three cuts $\tilde{C}_1, \tilde{C}_2, \tilde{C}_3$ (Fig. 355). Adding integrals as before, the integrals over the cuts cancel and the sum of the integrals over C_1 (counterclockwise) and C_2, C_3 (clockwise) is zero. Hence the integral over C_1 equals the sum of the integrals over C_2 and C_3 , all three now taken counterclockwise. Similarly for quadruply connected domains, and so on.

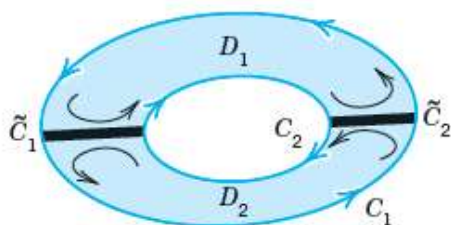


Fig. 354. Doubly connected domain

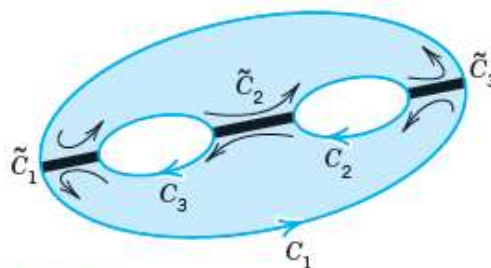


Fig. 355. Triply connected domain

14.3 Cauchy's Integral Formula

Cauchy's integral theorem leads to Cauchy's integral formula. This formula is useful for evaluating integrals as shown in this section. It has other important roles, such as in proving the surprising fact that analytic functions have derivatives of all orders, as shown in the next section, and in showing that all analytic functions have a Taylor series representation (to be seen in Sec. 15.4).

THEOREM 1

Cauchy's Integral Formula

Let $f(z)$ be analytic in a simply connected domain D . Then for any point z_0 in D and any simple closed path C in D that encloses z_0 (Fig. 356),

$$(1) \quad \oint_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0) \quad (\text{Cauchy's integral formula})$$

the integration being taken counterclockwise. Alternatively (for representing $f(z_0)$ by a contour integral, divide (1) by $2\pi i$),

$$(1^*) \quad f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz \quad (\text{Cauchy's integral formula}).$$

PROOF By addition and subtraction, $f(z) = f(z_0) + [f(z) - f(z_0)]$. Inserting this into (1) on the left and taking the constant factor $f(z_0)$ out from under the integral sign, we have

$$(2) \quad \oint_C \frac{f(z)}{z - z_0} dz = f(z_0) \oint_C \frac{dz}{z - z_0} + \oint_C \frac{f(z) - f(z_0)}{z - z_0} dz.$$

The first term on the right equals $f(z_0) \cdot 2\pi i$, which follows from Example 6 in Sec. 14.2 with $m = -1$. If we can show that the second integral on the right is zero, then it would prove the theorem. Indeed, we can. The integrand of the second integral is analytic, except

at z_0 . Hence, by (6) in Sec. 14.2, we can replace C by a small circle K of radius ρ and center z_0 (Fig. 357), without altering the value of the integral. Since $f(z)$ is analytic, it is continuous (Team Project 24, Sec. 13.3). Hence, an $\epsilon > 0$ being given, we can find a $\delta > 0$ such that $|f(z) - f(z_0)| < \epsilon$ for all z in the disk $|z - z_0| < \delta$. Choosing the radius ρ of K smaller than δ , we thus have the inequality

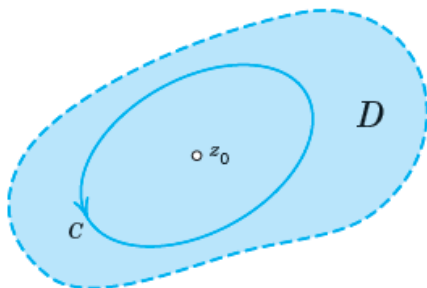


Fig. 356. Cauchy's integral formula

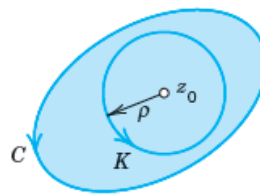


Fig. 357. Proof of Cauchy's integral formula

$$\left| \frac{f(z) - f(z_0)}{z - z_0} \right| < \frac{\epsilon}{\rho}$$

at each point of K . The length of K is $2\pi\rho$. Hence, by the ML -inequality in Sec. 14.1,

$$\left| \oint_K \frac{f(z) - f(z_0)}{z - z_0} dz \right| < \frac{\epsilon}{\rho} 2\pi\rho = 2\pi\epsilon.$$

Since $\epsilon (> 0)$ can be chosen arbitrarily small, it follows that the last integral in (2) must have the value zero, and the theorem is proved. ■

EXAMPLE 1 Cauchy's Integral Formula

$$\oint_C \frac{e^z}{z - 2} dz = 2\pi i e^z \Big|_{z=2} = 2\pi i e^2 = 46.4268i$$

for any contour enclosing $z_0 = 2$ (since e^z is entire), and zero for any contour for which $z_0 = 2$ lies outside (by Cauchy's integral theorem). ■

EXAMPLE 2 Cauchy's Integral Formula

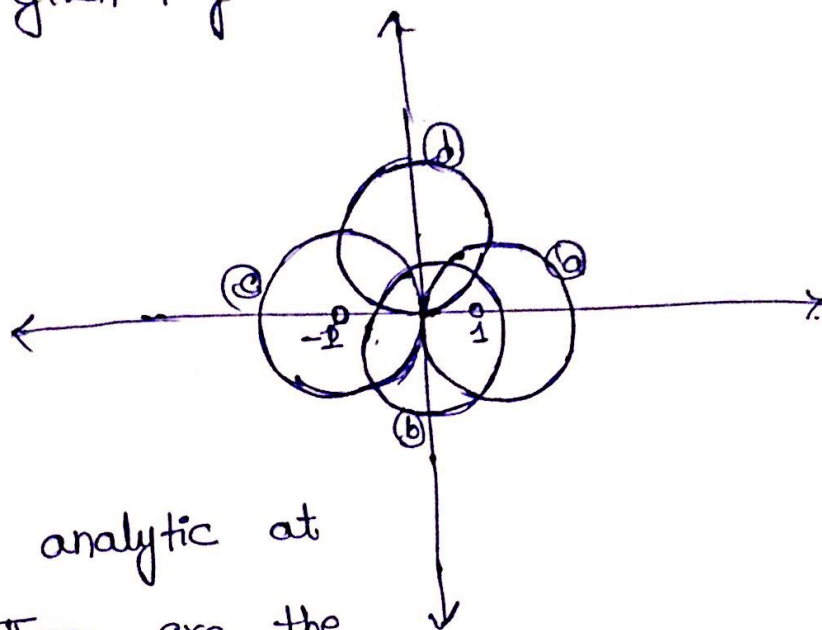
$$\begin{aligned} \oint_C \frac{z^3 - 6}{2z - i} dz &= \oint_C \frac{\frac{1}{2}z^3 - 3}{z - \frac{1}{2}i} dz \\ &= 2\pi i \left[\frac{1}{2}z^3 - 3 \right] \Big|_{z=i/2} \\ &= \frac{\pi}{8} - 6\pi i \quad (z_0 = \frac{1}{2}i \text{ inside } C). \end{aligned}$$

Integration Around Different Contours

Integrate

$$g(z) = \frac{z^2+1}{z^2-1} = \frac{z^2+1}{(z+1)(z-1)}$$

Counterclockwise around each of the four circles in given Fig



Solution

$g(z)$ is not analytic at -1 and 1 . These are the points we have to watch for. We consider each circle separately.

(a) The circle $|z-1|=1$ encloses the point $z_0=1$ where $g(z)$ is not analytic.

$$\oint \frac{\left(\frac{z^2+1}{z+1}\right)}{z-1} dz = 2\pi i \left(\frac{z^2+1}{z+1}\right)_{z=1}$$
$$= 2\pi i \left(\frac{2}{2}\right) = 2\pi i$$

Theorem 1

Cauchy's Integral Formula

$$\oint \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0)$$

b) gives the same as (a) by the principle of deformation of path.

c) The function $g(z)$ is as before, but $f(z)$ changes because we must take $z_0 = -1$ (instead of 1). This gives a factor

$$z - z_0 = z + 1$$

$$\text{so } g(z) = \left(\frac{z^2 + 1}{z - 1} \right) \cdot \frac{1}{z + 1}$$

$$\oint_C \frac{f(z) \cdot \frac{z^2 + 1}{z - 1}}{(z + 1)} dz = 2\pi i f(-1) \\ = 2\pi i \left[\frac{1+1}{-1-1} \right] = -2\pi i$$

d) gives 0,

$$\oint_C g(z) dz = 0$$

As path is closed
and $g(z)$ is analytic.

Multiply Connected domains

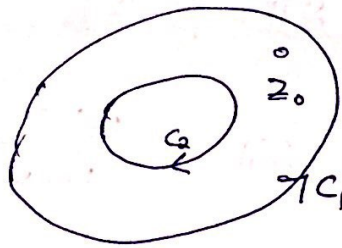
may be handled as in sec 14.2.

For instance, if $f(z)$ is analytic on C_1 and C_2 and in the ring-shaped domain bounded by C_1 and C_2

and z_0 is any point in that domain,
then

$$f(z_0) = \frac{1}{2\pi i} \oint_{C_1} \frac{f(z)}{z - z_0} dz + \frac{1}{2\pi i} \oint_{C_2} \frac{f(z)}{z - z_0} dz$$

where the outer integral (over C_1) is
taken counterclockwise and the inner
clockwise, as indicated in Fig.



Recommended Exercise

14.2, 14.3

Advanced Engineering Mathematics

by

Erwin Kreyszig.

